# MTH-418a: Inference-I

## 2022-2023: II Semester
## Mid Semester Examination

**Time Allowed: 120 Minutes** **Maximum Marks: 50**

**NOTE: (i) Start answer of every question on a new page. Moreover, attempt all the parts of a question at one place.**
**(ii) Answer each question legibly, clearly and concisely. Illegible answers will not be graded.**
**(ii) For each question, provide details of all the steps involved in arriving at the final answer(s)/conclusion(s) and box your final answer(s)/conclusion(s).**

1. Let $X_1, X_2$ be a random sample from a population having the p.d.f.

$$f_\theta(x) = \begin{cases} (x-\theta)e^{-(x-\theta)}, & \text{if } x \geq \theta \\ 0, & \text{otherwise} \end{cases},$$

where $\theta \in \mathbb{R} = \Theta$, say, is an unknown parameter. $\boxed{\text{3+2+2+2+2+3=14 Marks}}$

(i) Find a minimal sufficient statistic for $\theta$;

(ii) Is $\underline{T} = (X_1 + X_2, |X_1 - X_2|)$ a minimal sufficient statistic for $\theta$;

(iii) Is $\underline{U} = (X_1 + X_2, X_1 - X_2)$ a sufficient statistic for $\theta$;

(iv) Is $\underline{T} = (X_1 + X_2, |X_1 - X_2|)$ a complete statistic?

(v) Find the MME of $\theta$;

(vi) Find the MLE of $\theta$.

2. Let $X_1, \ldots, X_n$ be a random sample from $\mathrm{U}(0, \theta)$, where $\theta \in (0, \infty) = \Theta$, say, is an unknown parameter. For a fixed $t \in (0, \theta)$, using the Basu Theorem, find the conditional p.d.f. of $X_1$, given $X_{(n)} = t$. Hence, find the conditional expectation $E(X_1 | X_{(n)})$. $\boxed{\text{6+2=8 Marks}}$

3. (a) Let $X_1, \ldots, X_n$ be a random sample from $\mathrm{N}(\mu, \sigma^2)$, where $\underline{\theta} = (\mu, \sigma) \in \mathbb{R} \times (0, \infty) = \Theta$, say. Assuming that both $\mu$ and $\sigma$ are unknown, find the UMVUE of $\psi(\underline{\theta}) = \mu^3$. $\boxed{\text{4 Marks}}$

(b) Let $X_1, \ldots, X_n$ be a random sample from Poisson$(\theta)$, where $\theta \in (0, \infty) = \Theta$, say, is an unknown parameter. Find the UMVUE of $\psi(\theta) = \theta e^\theta$. $\boxed{\text{4 Marks}}$

4. Let $X_1, X_2$ be a random sample from a population having the p.d.f.

$$f_\theta(x) = \begin{cases} \frac{1}{\theta} e^{-\frac{x}{\theta}}, & \text{if } x > 0 \\ 0, & \text{otherwise} \end{cases},$$

where $\theta \in (0, \infty) = \Theta$, say, is an unknown parameter. Consider estimation of $\theta$, under the loss function $L(\theta, d)$, that is convex in $d \in \mathbb{R}$, for every fixed $\theta \in \Theta$. Let $\delta$ be a randomized estimator for $\theta$, such that, for any fixed $\underline{x} = (x_1, x_2) \in (0, \infty)^2$, $\delta(x_1, x_2) \sim p_{\underline{x}}$, where

$$p_{\underline{x}}(a) = \begin{cases} \frac{2}{x_1} e^{-\frac{2a}{x_1}}, & \text{if } a > 0 \\ 0, & \text{otherwise} \end{cases},$$

Let $T = X_1 + X_2$.

(i) Find a randomized estimator $\delta_1(T)$, based on $T$, that has the same risk function as the randomized estimator $\delta$, defined above; $\boxed{\text{4 Marks}}$

(ii) Find a non-randomized estimator $\delta_2(X_1)$, based on $X_1$ alone, that dominates the randomized estimator $\delta_1(T)$, derived in (a) above; $\boxed{\text{2 Marks}}$

(iii) Find a non-randomized estimator $\delta_3(\overline{X})$, based on $\overline{X} = \frac{X_1 + X_2}{2}$, that dominates the randomized estimator $\delta(X_1, X_2)$ $\boxed{\text{4 Marks}}$

5. Let $X$ be a r.v. with the p.m.f.

$$f_\theta(k) = \begin{cases} 2\theta(1-\theta), & \text{if } k = -1 \\ \theta^k (1-\theta)^{3-k}, & \text{if } k \in \{0, 1, 2, 3\} \\ 0, & \text{otherwise} \end{cases},$$

where $\theta \in (0, 1) = \Theta$ is unknown.

(i) Find the class of all unbiased estimators of $\psi(\theta) = \theta$; $\boxed{\text{5 Marks}}$

(ii) Find the UMVUE of $\theta$, provided it exists. $\boxed{\text{5 Marks}}$

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Model Solutions

Problem No. 1

(i) For $\underline{x}, \underline{y} \in \mathcal{X}$

$$\frac{f_\theta(\underline{x})}{f_\theta(\underline{y})} = \frac{(\theta - x_{(1)})(\theta - x_{(2)})\, I(\theta \le x_{(1)})}{(\theta - y_{(1)})(\theta - y_{(2)})\, I(\theta \le y_{(1)})}$$

is independent of $\theta \in \Theta = \mathbb{R}$

$\Leftrightarrow (x_{(1)}, x_{(2)}) = (y_{(1)}, y_{(2)})$

$\Leftrightarrow (X_{(1)}, X_{(2)})$ is minimal sufficient for $\theta$. ... 3 MARKS

(ii) $T = (X_1 + X_2, |X_1 - X_2|) = (X_{(1)} + X_{(2)}, X_{(2)} - X_{(1)})$ is a 1-1 function of minimal sufficient statistic $(X_{(1)}, X_{(2)})$

... 2 MARKS

$\Rightarrow T$ is minimal sufficient for $\theta$.

(iii) Minimal sufficient statistic $(X_{(1)}, X_{(2)})$ is a function of $\underline{U} = (X_1 + X_2, X_1 - X_2)$

$\Rightarrow \underline{U}$ is sufficient for $\theta$ ... 2 MARKS

(iv) $T = (X_1 + X_2, X_{(2)} - X_{(1)})$

$\psi(T) = X_{(2)} - X_{(1)}$ is ancillary for $\theta$

$\Rightarrow E_\theta[X_{(2)} - X_{(1)}] = c, \ \forall \theta \in \Theta$

where $c$ is some constant that does not depend on $\theta$

$\Rightarrow E_\theta[X_{(2)} - X_{(1)} - c] = 0, \ \forall \theta \in \Theta$, i.e. $E_\theta(\psi(T) - c) = 0 \ \forall \theta \in \Theta$

But $P_\theta(X_{(2)} - X_{(1)} - c) = 0, \ \forall \theta \in \Theta$

i.e. $P_\theta(\psi(T) - c) = 0, \ \forall \theta \in \Theta$

$\Rightarrow T$ is not complete $\Rightarrow T$ is not complete-sufficient ... 2 MARKS

(V) $E_\theta(X_1) = \theta + 2, \quad \forall\, \theta \in \Theta$

$\Rightarrow \hat{\theta}_{MME} = \bar{X} - 2$ ... $\boxed{2 \text{ MARKS}}$

(VI) For fixed $\underline{x} \in \mathcal{X}$,

$$L_{\underline{x}}(\theta) = (x_{(1)} - \theta)(x_{(2)} - \theta)\, e^{-(x_{(1)} + x_{(2)})}\, e^{2\theta}, \quad \theta \in (-\infty, x_{(1)})$$

$$l_{\underline{x}}(\theta) = \ln L_{\underline{x}}(\theta)$$

$$= \ln(x_{(1)} - \theta) + \ln(x_{(2)} - \theta) + 2\theta - (x_{(1)} + x_{(2)}), \quad \theta \le x_{(1)}$$

$$\frac{\partial}{\partial\theta} l_{\underline{x}}(\theta) = -\frac{1}{x_{(1)} - \theta} - \frac{1}{x_{(2)} - \theta} + 2 \;>(<)\; 0$$

$$\Leftrightarrow k_{\underline{x}}(\theta) = \theta^2 - (x_{(1)} + x_{(2)} - 1)\theta + x_{(1)} x_{(2)} - \frac{x_{(1)} + x_{(2)}}{2} \;>(<)\; 0, \quad \theta \le x_{(1)}$$

$$\frac{\partial}{\partial\theta} k_{\underline{x}}(\theta) = 2\theta - (x_{(1)} + x_{(2)} - 1) \;>(<)\; 0$$

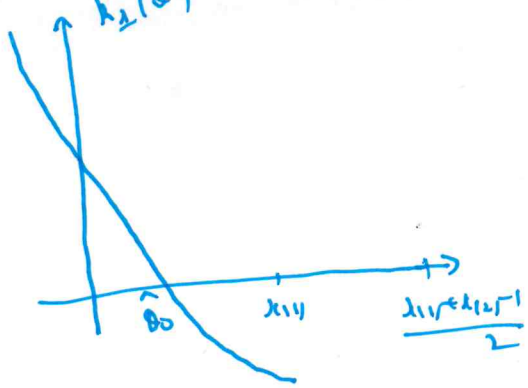
$$\Leftrightarrow \theta \;>(<)\; \frac{x_{(1)} + x_{(2)} - 1}{2}$$

<u>Case I</u>: $x_{(1)} \le x_{(2)} - 1 \quad \left(\Leftrightarrow x_{(1)} \le \frac{x_{(1)} + x_{(2)} - 1}{2}\right)$

$$k_{\underline{x}}(x_{(1)}) = x_{(1)}^2 - (x_{(1)} + x_{(2)} - 1)\, x_{(1)} + x_{(1)} x_{(2)} - \frac{x_{(1)} + x_{(2)}}{2}$$

$$= \frac{x_{(1)} - x_{(2)}}{2} < 0 \quad ; \quad \lim_{\theta \to -\infty} k_{\underline{x}}(\theta) = \infty$$

Let $\hat{\theta}_0 = \hat{\theta}_0(x_{(1)}, x_{(2)}) \; (\le x_{(1)})$ be the unique root of $k_{\underline{x}}(\theta) = 0$




$k_{\underline{x}}(\theta) \;>(<)\; 0 \Leftrightarrow \theta < \hat{\theta}_0 \; (\theta > \hat{\theta}_0)$

$\Rightarrow \frac{\partial}{\partial\theta} l_{\underline{x}}(\theta) \;>(<)\; 0 \Leftrightarrow \theta < \hat{\theta}_0 \; (\theta > \hat{\theta}_0)$

$\Rightarrow \hat{\theta}_0(x_{(1)}, x_{(2)})$ is the MLE of $\theta$

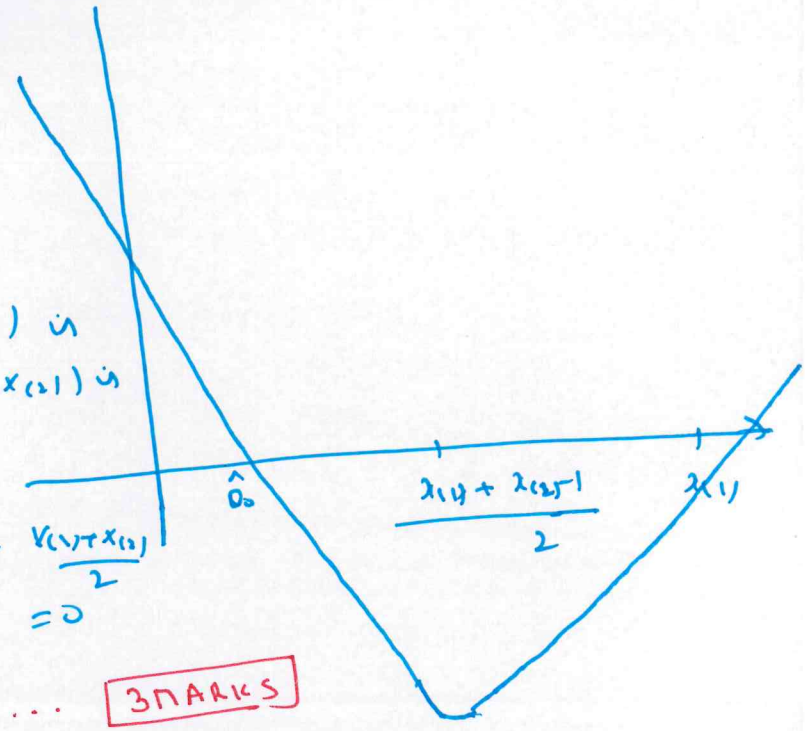
Case II $\quad x_{(1)} \geq x_{(2)} - 1 \quad \left( \Leftrightarrow \quad x_{(1)} \geq \dfrac{x_{(1)} + x_{(2)} - 1}{2} \right)$

Note the $k_{\underline{x}}(\theta) = \dfrac{x_{(1)} - x_{(2)}}{2} < 0$ and $\lim_{\theta \to -\infty} k_{\underline{x}}(\theta) = \infty$

$k_{\underline{x}}(\theta) > (<) \, 0 \Leftrightarrow \theta < (>) \, \hat{\theta}_0$

$\Rightarrow \dfrac{\partial}{\partial \theta} l_{\underline{x}}(\theta) > (<) \, 0 \Leftrightarrow \theta < (>) \, \hat{\theta}_0$

$\Rightarrow \hat{\theta}_0 \equiv \hat{\theta}_0(x_{(1)}, x_{(2)})$ is the MLE




Thus in either case $\hat{\theta}_0(x_{(1)}, x_{(2)})$ is the MLE of $\theta$, where $\hat{\theta}_0(x_{(1)}, x_{(2)})$ is the unique root of the equation

$$\theta^2 - (x_{(1)} + x_{(2)} - 1)\theta + x_{(1)} x_{(2)} - \frac{x_{(1)} + x_{(2)}}{2} = 0$$

in the interval $(-\infty, x_{(1)}]$.

... $\boxed{\text{3 MARKS}}$

Alter $\quad L_{\underline{x}}(\theta) = (x_{(1)} - \theta)(x_{(2)} - \theta) \, e^{-(x_{(1)} + x_{(2)})} \, e^{2\theta}, \quad \theta \leq x_{(1)}$

$L_{\underline{x}}(x_{(1)}) = 0 \qquad \lim_{\theta \to -\infty} L_{\underline{x}}(\theta) = 0, \qquad L_{\underline{x}}(\theta) \geq 0 \quad \forall \, \theta \leq x_{(1)}$

and $L_{\underline{x}}(\theta)$ is bounded above (i.e. maximum exists)

Since at boundary $\theta = x_{(1)}$ and $\theta \to -\infty$, $L_{\underline{x}}(\cdot)$ achieves its minimum, $L_{\underline{x}}(\theta)$ achieves its maximum at its critical point

$$\frac{\partial}{\partial \theta} L_{\underline{x}}(\theta) = 0$$

i.e. $\theta^2 - (x_{(1)} + x_{(2)} - 1)\theta + x_{(1)} x_{(2)} - \dfrac{x_{(1)} + x_{(2)}}{2} = 0$

... $\boxed{\text{3 MARKS}}$

**Problem No. 2** For fixed $t>0$, and $x \in (-\infty, \infty)$.

$$P_\theta(X_1 \le x \mid X_{(n)} = t) = P_\theta\left(\frac{X_1}{X_{(n)}} \le \frac{x}{t} \,\middle|\, X_{(n)} = t\right).$$

Clearly, for $x<0$, $P_\theta(X_1 \le x \mid X_{(n)}=t) = 0$ and, for $x \ge t$, $P_\theta(X_1 \le x \mid X_{(n)}=t) = 1$. Now $X_{(n)}$ is a complete-sufficient statistic and $\frac{X_1}{X_{(n)}}$ is an ancillary statistic, so by the Basu theorem, $\frac{X_1}{X_{(n)}}$ and $X_{(n)}$ are statistically independent. Thus, for fixed $t \ge 0$ and $0 \le x < t$,

$$P_\theta(X_1 \le x \mid X_{(n)} = t) = P_\theta\left(\frac{X_1}{X_{(n)}} \le \frac{x}{t}\right)$$

$$= 1 - P_\theta\left(X_{(n)} < \frac{t}{x} X_1\right)$$

$$= \begin{cases} 0, & x<0 \\ 1 - P\left(X_j < \frac{t}{x} X_1,\ j=2,\dots,n\right), & 0 \le x < t \\ 1 & x \ge t \end{cases}$$

WLOG we may take $\theta = 1$, as $X_1/X_{(n)}$ is ancillary.

For fix $t>0$ and $0 \le x < t$,

$$P\left(X_j < \frac{t}{x} X_1,\ j=2,\dots,n\right) = \int_0^1 \prod_{j=2}^{n} P\left(X_j < \frac{t}{x} z\right) dz$$

$$= \int_0^{x/t} \left(\frac{t}{x} z\right)^{n-1} dz + \int_{x/t}^1 dz$$

$$= \frac{1}{n}\left(\frac{x}{t}\right) + 1 - \frac{x}{t}$$

$$= 1 - \frac{n-1}{n}\frac{x}{t}$$

$$\Rightarrow P_\theta(X_1 \le x \mid X_{(n)} = t) = \begin{cases} 0 & x<0 \\ \frac{n-1}{n}\frac{x}{t} & 0 \le x < t \\ 1 & x \ge t \end{cases}$$

... 3 MARKS

$$P_0(X_1 = t \mid X_{(n)} = t) = P(X_1 \le t \mid X_{(n)} = t) - P(X_1 < t \mid X_{(n)} = t)$$

$$= 1 - \frac{n-1}{n} = \frac{1}{n}$$

Then the conditional p.d.f. of $X_1$ given $X_{(n)} = t$ is (for fixed $t > 0$)

$$g(x) = \begin{cases} \frac{1}{n} & x = t \\ \frac{n-1}{n}\frac{1}{t} & 0 < x < t \\ 0 & \text{o.w.} \end{cases}$$ ... $\boxed{\text{3 MARKS}}$

$$E(X_1 \mid X_{(n)} = t) = \frac{1}{n} \times t + \int_0^t x \times \frac{n-1}{n}\frac{1}{t}\, dx$$

$$= \frac{t}{n} + \frac{n-1}{2n} t = \frac{n+1}{2n} t$$

$$E(X_1 \mid X_{(n)}) = \frac{n+1}{2n} X_{(n)}$$ ... $\boxed{\text{2 MARKS}}$

Problem No. 3

(a) Let $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ and $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$. Then $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ and $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$ are independently distributed.

$$E_{\underline{\theta}}\left((\bar{X}-\mu)^3\right) = 0, \quad \forall\, \underline{\theta} \in \Theta$$

$$\Rightarrow E_{\underline{\theta}}(\bar{X}^3) - 3\mu\left(\frac{\sigma^2}{n} + \mu^2\right) + 3\mu^3 - \mu^3 = 0, \quad \forall\, \underline{\theta} \in \Theta$$

$$\Rightarrow E_{\underline{\theta}}(\bar{X}^3) - \frac{3}{n}\mu\sigma^2 = \mu^3, \quad \forall\, \underline{\theta} \in \Theta \quad \ldots \boxed{2\text{MARKS}}$$

$$E_{\underline{\theta}}(\bar{X}S^2) = E_{\underline{\theta}}(\bar{X})\,E_{\underline{\theta}}(S^2) = \mu\sigma^2$$

$$\Rightarrow E_{\underline{\theta}}\left(\bar{X}^3 - \frac{3}{n}\bar{X}S^2\right) = \mu^3, \quad \forall\, \underline{\theta} \in \Theta$$

Since $(\bar{X}, S^2)$ is a complete-sufficient statistic, by the Lehmann-Scheffe theorem $\delta_0(\underline{X}) = \bar{X}^3 - \frac{3}{n}\bar{X}S^2$ is the UMVUE of $\psi(\underline{\theta}) = \mu^3$. $\ldots \boxed{2\text{MARKS}}$

(b) $T = \sum_{i=1}^{n} X_i$ is a complete-sufficient statistic. Also $T \sim \text{Poisson}(n\theta)$, $\theta > 0$.

By the Lehmann-Scheffe theorem, the UMVUE of $\psi(\theta) = \theta e^{\theta}$ is $\delta_1(T)$, where

$$E_\theta(\delta_1(T)) = \theta e^{\theta}, \quad \forall\, \theta > 0$$

$$\Leftrightarrow \sum_{k=0}^{\infty} \delta_1(k)\, \frac{e^{-n\theta}(n\theta)^k}{\lfloor k} = \theta e^{\theta}, \quad \forall\, \theta > 0 \quad \ldots \boxed{2\text{MARKS}}$$

$$\Leftrightarrow \sum_{k=0}^{\infty} \delta_1(k)\, \frac{n^k\theta^k}{\lfloor k} = \theta e^{(n+1)\theta}, \quad \forall\, \theta > 0$$

$$\Leftrightarrow \sum_{k=0}^{\infty} \delta_1(k)\, \frac{n^k\theta^k}{\lfloor k} = \theta \sum_{k=0}^{\infty} \frac{(n+1)^k\theta^k}{\lfloor k}, \quad \forall\, \theta > 0$$

$$\sum_{k=0}^{\infty} \delta_1(k)\, \frac{n^k\theta^k}{\lfloor k} = \sum_{k=1}^{\infty} \frac{(n+1)^{k-1}\theta^k}{\lfloor k-1}, \quad \forall\, \theta > 0$$

$$\Leftrightarrow \delta_1(0) = 0, \quad \delta_1(k) = \frac{k}{n+1}\left(\frac{n+1}{n}\right)^k, \quad k = 1, 2, \ldots$$

$$\Rightarrow \delta_1(T) = \frac{T}{n+1}\left(\frac{n+1}{n}\right)^T \text{ is the UMVUE of } \psi(\theta) = \theta e^{\theta}. \quad \ldots \boxed{2\text{MARKS}}$$

**Problem No. 4** $T = X_1 + X_2$ is a Complete-Sufficient statistic

Also $U = \frac{X_1}{T}$ is ancillary. and $U \sim \text{Beta}(1, 1) \equiv U(0,1)$

(i) The derived estimator is $\delta_1(T)$, where for fixed $t \in (0, a)$, $\delta_1(t) \sim p_t(a)$, $a > 0$, $t > 0$

$$p_t(a) = E\left[ p_{\underline{X}}(a) \mid T = t \right]$$

$$= \begin{cases} E\left[ \frac{2}{X_1} e^{-\frac{2a}{X_1}} \mid T = t \right], & a > 0 \\ 0, & \text{o.w.} \end{cases} \quad \ldots \boxed{\text{2 MARKS}}$$

For $a > 0$ and $t > 0$,

$$E\left[ \frac{2}{X_1} e^{-\frac{2a}{X_1}} \mid T = t \right] = \frac{2}{t} E\left[ \frac{1}{U} e^{-\frac{2a}{tU}} \right]$$

$$= \frac{2}{t} \int_0^1 \frac{1}{u} e^{-\frac{2a}{tu}} \, du \quad \ldots \boxed{\text{2 MARKS}}$$

$$= \frac{2}{t} \int_{\frac{2a}{t}}^{\infty} \frac{e^{-z}}{z} \, dz$$

(ii) The derived estimator is

$$\delta_2(X_1) = E\left[ Y_{X_1} \right], \quad \text{where for fixed } x \ \ Y_x \sim \text{Exp}\left(\tfrac{x}{2}\right)$$

$$= \frac{X_1}{2} \quad \ldots \boxed{\text{2 MARKS}}$$

(iii) The derived estimator is

$$\delta_2(\bar{X}) = E\left[ \delta_2(X_1) \mid \bar{X} \right]$$

$$= \frac{1}{2} E\left[ X_1 \mid \bar{X} \right] = \frac{\bar{X}}{2} E\left[ \frac{X_1}{\bar{X}} \mid \bar{X} \right] \quad \boxed{\text{2 MARKS}}$$

$$= \frac{\bar{X}}{2} E\left[ \frac{X_1}{\bar{X}} \right] \quad \text{(by Basu's thm, as } \frac{X_1}{\bar{X}} \text{ is ancillary and } \bar{X} \text{ is Complete-Suff)}$$

$$= \frac{\bar{X}}{2} E\left[ \text{Beta}(1, n-1) \right] \times n$$

$$= \frac{\bar{X}}{2} \quad \ldots \boxed{\text{2 MARKS}}$$

## Problem No. 5

(i) $\delta$ is an unbiased estimator of $\psi(\theta)=\theta$ iff

$$E_\theta(\delta(X)) = \theta, \quad \forall\, \theta \in (0,1)$$

$$\Leftrightarrow \quad 2\theta(1-\theta)\,\delta(-1) + (1-\theta)^3\,\delta(0) + \theta(1-\theta)^2\,\delta(1) + \theta^2(1-\theta)\,\delta(2) + \theta^3\,\delta(3) = \theta, \quad \forall\, \theta \in (0,1)$$

... $\boxed{\text{2 MARKS}}$

$$\Leftrightarrow \quad (\delta(3) - \delta(2) + \delta(1) - \delta(0))\,\theta^3 + (\delta(2) - 2\delta(1) + 3\delta(0) - 2\delta(-1))\,\theta^2 + (\delta(1) - 3\delta(0) + 2\delta(-1))\,\theta + \delta(0) = \theta, \quad \forall\, \theta \in (0,1)$$

$$\Leftrightarrow \quad \delta(0)=0, \quad \delta(1) + 2\delta(-1) = 1, \quad \delta(2) - 2\delta(1) - 2\delta(-1) = 0, \quad \delta(3) - \delta(2) + \delta(1) = 0$$

$$\Leftrightarrow \quad \delta(-1) = c, \quad \delta(0) = 0, \quad \delta(1) = 1-2c, \quad \delta(2) = 2-2c, \quad \delta(3) = 1$$

Thus the class of all unbiased estimators of $\theta$ is

$$D_0 = \{\delta_c : c \in \mathbb{R}\},$$

where $\delta_c(-1) = c$, $\delta_c(0) = 0$, $\delta_c(1) = 1-2c$, $\delta_c(2) = 2-2c$, $\delta_c(3) = 1$

... $\boxed{\text{3 MARKS}}$

(ii) $\delta_c \in D_0$ will be UMVUE for $\theta$ iff

$$E_\theta[\delta_c(X)\,U(X)] = 0, \quad \forall\, \theta \in (0,1) \text{ and } U \in \mathcal{U}$$

$U \in \mathcal{U} \Leftrightarrow \quad U(0)=0, \quad U(1) + 2U(-1) = 0, \quad U(2) - 2U(1) - 2U(-1) = 0,$

$$U(3) - U(2) + U(1) = 0$$

$$\Leftrightarrow \quad U(0)=0, \quad U(1) = -2U(-1), \quad U(2) = -2U(-1), \quad U(3) = 0 \quad \ldots (*)$$

.. $\boxed{\text{2 MARKS}}$

$$E_\theta[\delta_c(X)\,U(X)] = 0, \quad \forall\, \theta \in (0,1), \text{ and } U \in \mathcal{U}$$

$$\Leftrightarrow \quad \delta_c(0)U(0) = 0, \quad \delta_c(1)U(1) = -2\delta_c(-1)U(-1), \quad \delta_c(2)U(2) = -2\delta_c(-1)U(-1),$$

$$\delta_c(3)U(3) = 0 \quad \forall\, U \text{ satisfying } (*) \ldots (**)$$

$$\Leftrightarrow \quad \delta_c(1)U(1) = -2\delta_c(-1)U(-1), \quad \delta_c(2)U(2) = -2\delta_c(-1)U(-1)$$

<u>Case I</u> $U(-1) = 0$ and $U \in \mathcal{U}$

Here $U(k) = 0$, $\forall\, k = -1, 0, 1, 2, 3$ and and $\delta_c \in D_0$ would

~~work~~ satisfy $(**)$

<u>Case II</u>: $U(-1) = c \neq 0$

Here $U(1) \neq 0$, $U(2) \neq 0$, $\delta_c(1) = \delta_c(-1)$, $\delta_c(2) = \delta_c(-1)$

$\Rightarrow \delta_c(1) = \delta_c(2) \longrightarrow$ Contradiction

Thus no UMVUE exists

.. $\boxed{\text{3 MARKS}}$

$\boxed{8/8}$